# 200mA, 30V Dual Schottky Barrier Diode

## FEATURES

- AEC-Q101 qualified
- Fast switching speed
- Low forward voltage
- Surface mound device type
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

| KEY PARAMETERS | | |
|---|---|---|
| **PARAMETER** | **VALUE** | **UNIT** |
| $I_F$ | 200 | mA |
| $V_{RRM}$ | 30 | V |
| $I_{FSM}$ | 600 | mA |
| $T_{J\ MAX}$ | 125 | °C |
| $V_F$ at $I_F$=100mA | 800 | mV |
| Configuration | Dual die | |

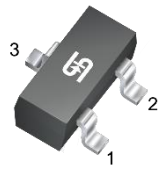
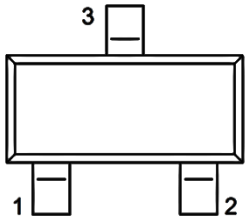
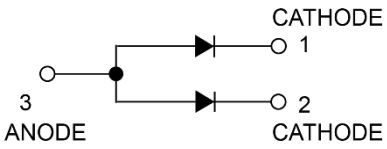
## APPLICATIONS

- Ultra high-speed switching
- Line termination
- Voltage clamping
- Reverse polarity protection

Pb-Free RoHS COMPLIANT HALOGEN FREE

## MECHANICAL DATA

- Case: SOT-23
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Weight: 8.00mg (approximately)

| PACKAGE: SOT-23 | PIN CONFIGURATION | CIRCUIT DIAGRAM |
|---|---|---|
|  |  |  |







## ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25°C unless otherwise noted)

| PARAMETER | | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|
| Power dissipation(1) | | $P_D$ | 200 | mW |
| Repetitive peak reverse voltage | | $V_{RRM}$ | 30 | V |
| Forward current | | $I_F$ | 200 | mA |
| Repetitive peak forward current | | $I_{FRM}$ | 300 | mA |
| Non-repetitive peak forward surge current | t < 1.0s | $I_{FSM}$ | 600 | mA |
| Junction temperature | | $T_J$ | -55 to +125 | °C |
| Storage temperature | | $T_{STG}$ | -55 to +150 | °C |

**Note:**

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

## THERMAL PERFORMANCE

| PARAMETER | SYMBOL | TYP | UNIT |
|---|---|---|---|
| Junction-to-ambient thermal resistance(1) | $R_{\Theta JA}$ | 500 | °C/W |

**Thermal Performance Note:**

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

## ELECTRICAL SPECIFICATIONS ($T_A$ = 25°C unless otherwise noted)

| PARAMETER | CONDITIONS | SYMBOL | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|
| Forward voltage(1) | $I_F$ = 0.1mA | $V_F$ | - | - | 240 | mV |
| | $I_F$ = 1mA | | - | - | 320 | |
| | $I_F$ = 10mA | | - | - | 400 | |
| | $I_F$ = 30mA | | - | - | 500 | |
| | $I_F$ = 100mA | | - | - | 800 | |
| Reverse breakdown voltage | $I_R$ = 100μA | $V_{BR}$ | 30 | - | - | V |
| Reverse current(2) | $V_R$ = 25V | $I_R$ | - | - | 2 | μA |
| Junction capacitance | f = 1MHz, $V_R$ = 1V | $C_J$ | - | - | 10 | pF |
| Reverse recovery time | $I_F$ = $I_R$ = 10mA, $I_{RR}$ = 1mA, $R_L$ = 100Ω | $t_{rr}$ | - | - | 5 | ns |

**Notes:**

1. Pulse test with PW=0.3 ms
2. Pulse test with PW=30 ms

## ORDERING INFORMATION

| ORDERING CODE | PACKAGE | PACKING |
|---|---|---|
| BAT54AH RFG | SOT-23 | 3,000 / 7” Tape & Reel |

## CHARACTERISTICS CURVES

($T_A$ = 25°C unless otherwise noted)

**Fig.1 Power Dissipation Curve**

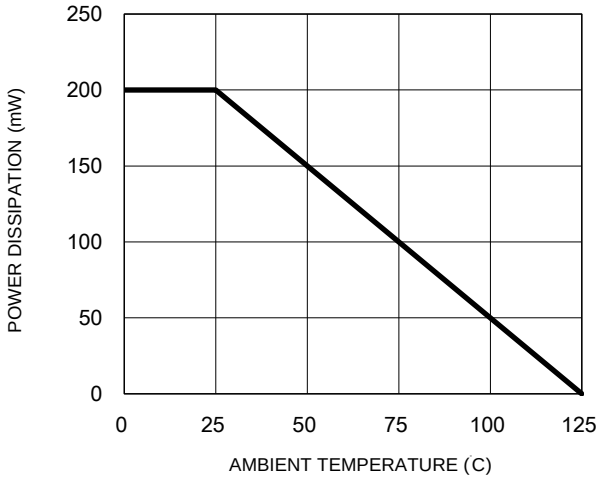



**Fig.2 Typical Junction Capacitance**

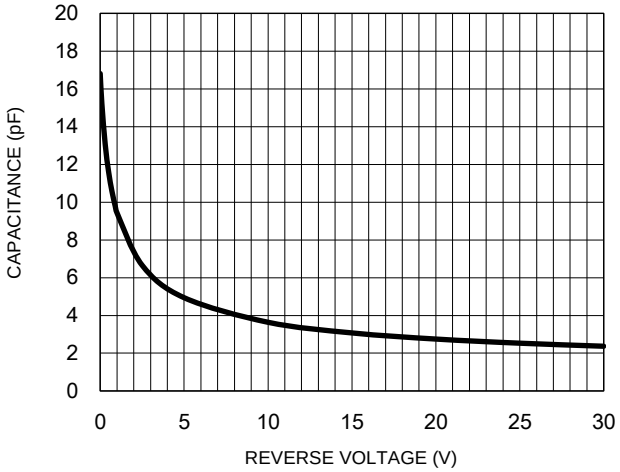



**Fig.3 Typical Reverse Characteristics**

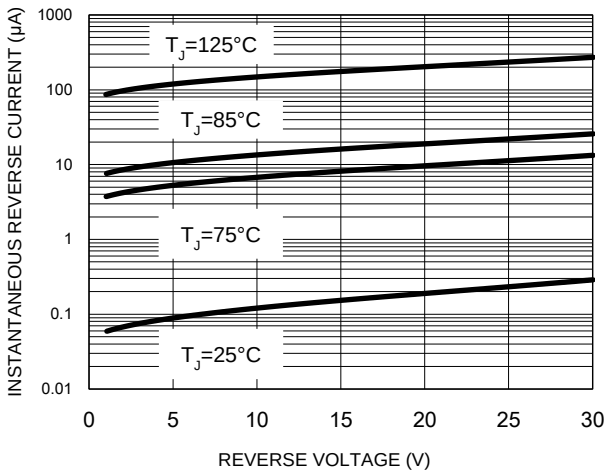
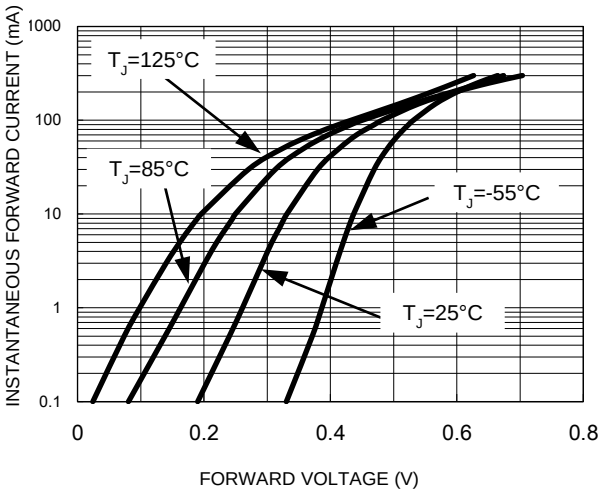
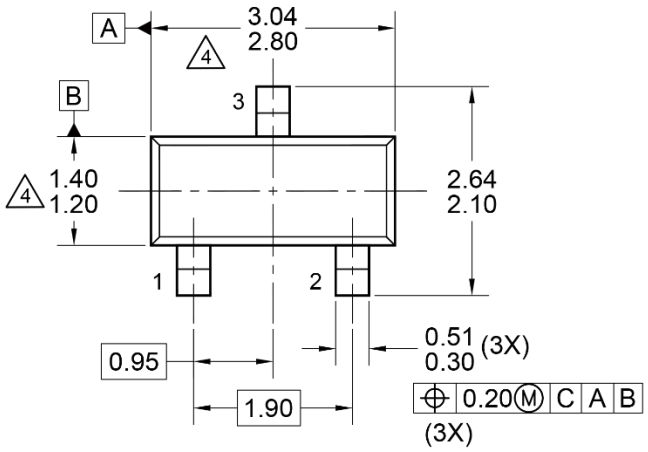


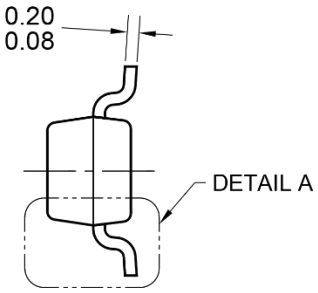

**Fig.4 Typical Forward Characteristics**

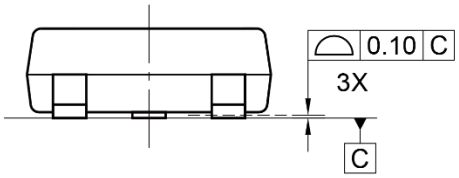


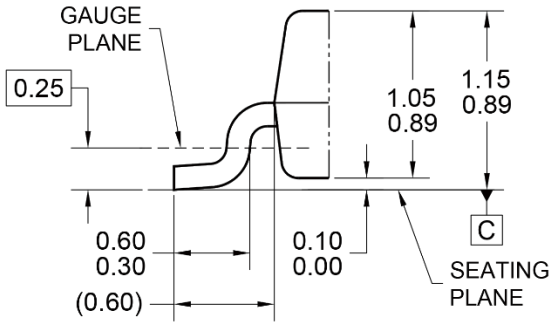
## PACKAGE OUTLINE DIMENSIONS

**SOT-23**

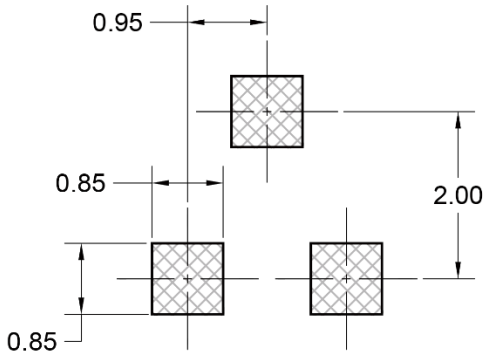


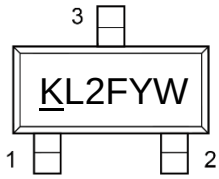




DETAIL A, ROTATED -90°
(SCALE 2:1)




SUGGESTED PAD LAYOUT




MARKING DIAGRAM

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: JEDEC TO-236, ISSUE H, VARIATION AA.
4. MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
5. DWG NO. REF: HQ2SD07-SOT23JEDEC-104 REV B.

KL2 = Device marking
F = Factory code
Y = Year code
W = Bi-Week code (A~Z)

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